Quantic

Web Application & Interface Design Project Supporting Documentation

Web Application & Interface Design

Vishant Prasad
12-22-2025

Quantic MSSE Web Application & Interface Design Project

Café Fausse Presentation Project

Contents

Supporting Documentation 2
AI Documentation 2
How AI was Used 2
Code Breakdown 2
Learning Outcomes 2
Cross-Platform Testing 2
Toolchain 2
Devices Selected for Testing 2
Testing Scope 3
Testing Checklist 3
Requirements Traceability Matrix 3

Supporting Documentation

AI Documentation

Tools Used: Claude (Anthropic) – AI Assistant for Code Generation & Guidance (Sonnet 4.5).

How AI was Used

1. **Project Structure**: Claude provided recommendations for React folder structure.
2. **Component Generation**: All React components were generated with Claude assistance.
3. **Backend Code**: Flask app, database models and API endpoints were generated with review and assistance from Claude AI model.
4. **CSS Styling**: All CSS files were created with Claude assistance and guidance.
5. **Problem Solving**: Debugging assistance for PostgreSQL setup, CORS issues, proxy configuration.

Code Breakdown

- **AI-Generate**: Evaluated by the AI model that ~90% of initial code structure, components and backend where generated by Claude AI.
- **Manually Modified**: Configuration files, image paths, bug fixes, customizations, code layout for readability and formatting and some aspects of the website frontend styling.
- **Human Decision-Making**: Architecture choices, feature prioritization, testing strategy.

Learning Outcomes

Through AI assistance, learned about:

- React Router implementation
- Flask-React integration
- PostgreSQL database design
- CORS and proxy configuration
- Responsive CSS design patterns

Cross-Platform Testing

Toolchain

DevTools: Opened the website on a web browser – used Google Chrome for all the testing and staging in this project at the address; `http://localhost:3000`. Used **F12 Inspect** and toggled the device using the **Device Toolbar** to view the page in the perspective of a mobile device.

Also used the Mobile Simulator - Responsive Testing Tool to view and test the website responsiveness and UX from the perspective of mobile users.

Devices Selected for Testing

The following is a list of devices that were able to be selected via a combination of the above tools to view the website through the lens of:

- iPhone 12 Pro (390 x 844)
- iPhone SE (375 x 667)
- Pixel 5 (393 x 851)
- iPad Air (820 x 1180)
- Samsung Galaxy S20 Ultra (412 x 915)
- 320px - Small Phones (iPhone SE)

- 375px - Medium Phones (iPhone 12/13 Mini)
- 414px - Large Phones (iPhone 12 Pro Max)
- 768px - Tablets (iPad)
- 1024px - Large Tablets/Small Laptops

Testing Scope

- **Rotate Device**: Click the rotate icon to switch portrait/landscape.
- **Network Throttling**: Simulate 3G/4G speeds.
- **Touch Simulation**: Using a mouse to simulate touch events.

Testing Checklist

Generated a checklist for user testing the cross-platform views:

Navigation

- Does the hamburger menu appear?
- Does it open/close properly?
- Are links in the navigation hamburger menu clickable?

Home Page

- Hero image displays correctly?
- Text is readable?
- Cards stack vertically?

Menu Page

- Menu items are readable?
- Prices align properly?

Reservations Page

- Form fields are accessible?
- Date/time pickers work on mobile?
- Submit button is visible?

Gallery Page

- Images display in grid?
- Lightbox works on mobile?
- Touch/tap works for image selection?

Footer

- Newsletter signup form fits?
- Contact information is readable?

Requirements Traceability Matrix

Attached to the end of this document is an Excel table document exported Requirements Traceability Matrix used for tracking the features and development activities implemented and performed during the project sprint.

SEE END OF DOCUMENT MERGED A3 TABLE